

PSP - Aurora Postgres FY24 Peak Readiness

- Monolith
 - pg_cluster_size
 - pg_catalog_size
 - pg_db_settings
 - pg_vacuum_settings
 - pg_vacuum_efficiency
- Audit
 - pg_cluster_size
 - pg_catalog_size
 - pg_db_settings
 - pg_vacuum_settings
 - pg_vacuum_efficiency

Monolith

pg_cluster_size

```
pspapg02=> SELECT round(sum(pg_database_size(datname))/1024/1024/1024) AS db_size_gb
pspapg02-> FROM pg_database;
db_size_gb
-----
      29434
(1 row)
```

pg_catalog_size

```
pspapg02=> SELECT nspname ,
pspapg02-> pg_size_pretty(sum(pg_relation_size(c.oid))) AS size
pspapg02-> FROM pg_class c
pspapg02-> LEFT JOIN pg_namespace n ON (n.oid = c.relnamespace)
pspapg02-> WHERE nspname IN ('pg_catalog', 'information_schema')
pspapg02-> GROUP by nspname
pspapg02-> ORDER BY nspname DESC
pspapg02-> ;
nspname | size
-----+-----
pg_catalog | 199 MB
information_schema | 88 kB
(2 rows)
```

pg_db_settings

```
pspapg02=> SELECT name, setting
pspapg02-> FROM pg_settings
pspapg02-> WHERE name IN ('auto_explain.log_analyze', 'auto_explain.log_buffers', 'auto_explain.log_timing', 'auto_explain.log_min_duration', 'random_page_cost', 'log_statement', 'shared_preload_libraries', 'work_memory', 'idle_in_transaction_session_timeout') order by 1;
name | setting
-----+-----
auto_explain.log_analyze | on
auto_explain.log_buffers | on
auto_explain.log_min_duration | 2000
auto_explain.log_timing | on
idle_in_transaction_session_timeout | 1800000
log_statement | ddl
random_page_cost | 1
shared_preload_libraries | rdsutils,pg_stat_statements,auto_explain,pg_hint_plan,apg_plan_mgmt
(8 rows)
```

pg_vacuum_settings

```
pspag02=> -- auto vacuum setting
pspag02=> select name, setting, source
pspag02->   from pg_settings
pspag02->   where name in ('autovacuum', 'autovacuum_freeze_max_age');
      name          | setting | source
-----+-----+-----
 autovacuum         | on      | configuration file
 autovacuum_freeze_max_age | 500000000 | configuration file
(2 rows)

pspag02=> -- current max age
pspag02=> select max(age(datfrozenxid))
pspag02->   from pg_database;
      max
-----
244092653
(1 row)
```

pg_vacuum_efficiency

```
pspag02=> SELECT current_database() shard_id, nspname, relname, relpages tot_pages, all_visible visible_pages,
all_visible/relpages::float visibility_factor
pspag02-> FROM (SELECT relname,relpages,nspname, (select all_visible from pg_visibility_map_summary
((nspname||'.'||relname)::regclass) )
pspag02(>   FROM pg_class c LEFT JOIN
pspag02(>   pg_namespace n ON (n.oid = c.relnamespace)
pspag02(>   WHERE relkind = 'r'
pspag02(>   ) x
pspag02-> WHERE all_visible > 0
pspag02-> AND all_visible/relpages::float <= 0.96
pspag02-> ORDER BY visibility_factor desc ;
 shard_id | nspname | relname | tot_pages | visible_pages |
visibility_factor
-----+-----+-----+-----+-----+
+-----+
pspag02 | pspadm | psp_gems_upload_batch | 124 | 119 |
0.9596774193548387
pspag02 | pspadm | psp_employee_bank_account | 1801477 | 1727288 |
0.958817681269314
pspag02 | pspadm | psp_company_event_p4 | 6535684 | 6265123 |
0.9586024966935366
pspag02 | pspadm | psp_law_rate_range | 24 | 23 |
0.9583333333333334
pspag02 | pspadm | psp_bill | 719877 | 689226 |
0.9574218929067049
pspag02 | pspadm | psp_gems_monthly_balance | 185 | 177 |
0.9567567567567568
pspag02 | pspadm | psp_on_hold_reason | 13546 | 12940 |
0.9552635464343717
pspag02 | pspadm | psp_nachafile | 973 | 929 |
0.9547790339157246
pspag02 | pspadm | psp_employee_law_qtr_totals | 11330567 | 10809858 |
0.9540438708848374
pspag02 | pg_catalog | pg_depend | 432 | 411 |
0.9513888888888888
pspag02 | pspadm | psp_company_pin | 47126 | 44717 |
0.9488817213427831
pspag02 | pspadm | psp_message_log | 37801 | 35817 |
0.9475146160154493
pspag02 | pspadm | psp_company_bank_account | 95907 | 90712 |
0.9458329423295484
pspag02 | pspadm | psp_companyagency_frmtemplate | 37964 | 35886 |
```

0.9452639342535033	pspagg02 pspadm	psp_bank_account	1939203	1828901
0.9431199312294793	pspagg02 pspadm	psp_auth_user_auth_role__assoc	461	433
0.9392624728850325	pspagg02 pspadm	psp_company_service	269058	252567
0.9387083825792208	pspagg02 pspadm	psp_qbdt_payroll_item_info	1489156	1395692
0.9372369315236282	pspagg02 pspadm	psp_company_law_rate	384774	360544
0.93702796966531	pspagg02 pspadm	psp_company_agency	63789	59743
0.936572136261738	pspagg02 pspadm	psp_state_report_output	2155	2015
0.9350348027842227	pspagg02 pspadm	psp_company_law	361987	336375
0.9292460778978251	pspagg02 pspadm	psp_agency_id_requirement	26	24
0.9230769230769231	pspagg02 pspadm	psp_company_payroll_item	676357	624231
0.9229312330618298	pspagg02 pspadm	psp_paycheck_usage_2023	3499784	3228052
0.9223574940624907	pspagg02 pg_catalog	pg_index	128	118
0.921875	pspagg02 pg_catalog	pg_inherits	25	23
0.92	pspagg02 pspadm	psp_tax_payment_on_hold_reason	12048	11069
0.9187416998671979	pspagg02 pspadm	psp_fset_filing_detail	4109	3773
0.9182282793867121	pspagg02 pspadm	psp_fset_file	24	22
0.9166666666666666	pspagg02 pg_catalog	pg_constraint	198	181
0.9141414141414141	pspagg02 pspadm	psp_gems_ledger_posting_rule	23	21
0.9130434782608695	pspagg02 pspadm	psp_ddcompany_service_info	14878	13422
0.902137384056997	pspagg02 pspadm	psp_ee_payrollitem_qtrtotals	629175	566865
0.9009655501251639	pspagg02 pspadm	psp_fraud_bank_account	48877	43695
0.8939787630173701	pspagg02 pspadm	psp_rtbaautomationbackup	208	185
0.8894230769230769	pspagg02 perfstat	pg_stat_all_indexes_hist_p2023_11_07	5076	4512
0.8888888888888888	pspagg02 perfstat	pg_stat_all_tables_hist_p2023_11_07	2527	2245
0.888405223585279	pspagg02 pspadm	psp_auth_user	1794	1588
0.8851727982162765	pspagg02 perfstat	pg_stat_statements_hist_p2023_11_07	2355	2083
0.8845010615711253	pspagg02 pg_catalog	pg_statistic	2793	2465
0.882563551736484	pspagg02 pspadm	psp_entity_update_m112023	1789442	1567631
0.8760445993779067	pspagg02 pspadm	psp_address	2335752	2043343
0.8748116238367772	pspagg02 pspadm	psp_payee	132354	115426
0.8721005787509255	pspagg02 pspadm	psp_assisted_bundle_bill	1245	1083
0.8698795180722891	pspagg02 pspadm	psp_tax_company_service_info	1948	1676
0.8603696098562629	pspagg02 pspadm	psp_entity_update_m102023	2365186	2025990
0.8565880230983948	pspagg02 pg_catalog	pg_type	172	146
0.8488372093023255	pspagg02 pspadm	psp_employee_wage_plan	77232	65190
0.8440801740211311				

pspagg02 pspadm 0.8411404457325219	psp_employee_accrual		1555462		1308362	
pspagg02 pspadm 0.8381254233362948	psp_payroll_item_taxable_to		975879		817909	
pspagg02 pspadm 0.8348082595870207	psp_accounting_report_file		339		283	
pspagg02 perfstat 0.8331143232588699	host_stat_hist_default		3044		2536	
pspagg02 pg_catalog 0.8290598290598291	pg_attribute		1755		1455	
pspagg02 pg_catalog 0.8284789644012945	pg_class		309		256	
pspagg02 pspadm 0.8094906559179187	psp_comp_pmttemplate_pmtmethod		136450		110455	
pspagg02 pspadm 0.80746226352956	psp_tax_table_misc_data		10065793		8127748	
pspagg02 pg_catalog 0.7923076923076923	pg_proc		130		103	
pspagg02 perfstat 0.7863891592423449	pg_stat_all_indexes_hist_p2023_11_01		66785		52519	
pspagg02 perfstat 0.786380451206169	pg_stat_statements_hist_p2023_11_01		38386		30186	
pspagg02 pspadm 0.7711421098517872	psp_bpcompany_service_info		2294		1769	
pspagg02 perfstat 0.7687767204523513	pg_stat_statements_hist_p2023_11_02		36255		27872	
pspagg02 perfstat 0.7652465460431239	pg_stat_all_tables_hist_p2023_11_01		34598		26476	
pspagg02 perfstat 0.7630876068376068	pg_stat_all_tables_hist_p2023_11_02		33696		25713	
pspagg02 perfstat 0.7628576080039072	pg_stat_all_indexes_hist_p2023_10_31		67567		51544	
pspagg02 perfstat 0.7627457318327407	pg_stat_all_tables_hist_p2023_10_31		34031		25957	
pspagg02 perfstat 0.7601701320449931	pg_stat_all_indexes_hist_p2023_11_02		67477		51294	
pspagg02 pspadm 0.7306377414006492	psp_entitlement_message_2023		428230		312881	
pspagg02 perfstat 0.7265037593984962	pg_stat_activity_hist_p2023_11_02		1064		773	
pspagg02 pspadm 0.7217573221757322	psp_fraud_company		2868		2070	
pspagg02 pspadm 0.710623440588445	psp_employee_custom_field		1234440		877222	
pspagg02 pspadm 0.6692563817980022	psp_wc_paycheck_pending		901		603	
pspagg02 pspadm 0.6668770905751952	psp_companyagency_pmttemplate		121976		81343	
pspagg02 pspadm 0.6666666666666666	psp_user_setting		9		6	
pspagg02 pspadm 0.6658184902459712	psp_fraud_contact		3537		2355	
pspagg02 pspadm 0.6499845535990114	psp_fraud_address		3237		2104	
pspagg02 perfstat 0.5844638493334545	pg_stat_activity_sample_default		374018		218600	
pspagg02 pspadm 0.5586125788436872	psp_employee_payroll_item		3424973		1913233	
pspagg02 pspadm 0.5396564238161385	psp_employee_tax		7610772		4107202	
pspagg02 pspadm 0.5168981299031585	psp_company		951555		491857	
pspagg02 pspadm 0.4974344475352273	psp_individual		2192900		1090824	
pspagg02 pspadm 0.49278393452122266	psp_payee_bank_account		110448		54427	
pspagg02 pspadm 0.4913636835263026	psp_contact		327686		161013	
pspagg02 pspadm 0.40146406805153223	psp_qbdt_employee_info		2066434		829599	
pspagg02 pg_catalog	pg_foreign_table		28		11	

```

0.39285714285714285
pspag02 | pspadm | psp_employee | 3682094 | 1410524 |
0.3830765863120279
pspag02 | pspadm | psp_entitlement | 139266 | 39965 |
0.2869688222538164
pspag02 | pspadm | psp_entitlement_unit | 370440 | 77850 |
0.21015549076773565
pspag02 | pspadm | psp_pstub_employer_info | 41900 | 5869 |
0.14007159904534605
pspag02 | pspadm | psp_quickbooks_info | 256520 | 34287 |
0.13366209262435677
pspag02 | pspadm | psp_smssync_failure | 9 | 1 |
0.11111111111111111
pspag02 | perfstat | pg_stat_all_tables_hist_p2023_11_03 | 34211 | 1468 |
0.04291017508988337
pspag02 | perfstat | pg_stat_all_indexes_hist_p2023_11_03 | 71123 | 2950 |
0.04147744049041801
pspag02 | perfstat | pg_stat_statements_hist_p2023_11_03 | 53165 | 1309 |
0.024621461487820934
pspag02 | perfstat | pg_stat_activity_hist_p2023_11_03 | 3265 | 51 |
0.01562021439509954
pspag02 | ggschema | checkpoint | 22933 | 5 |
0.00021802642480268607
(96 rows)

```

Audit

pg_cluster_size

```

prodapgib=> SELECT round(sum(pg_database_size(datname))/1024/1024/1024) AS db_size_gb
prodapgib-> FROM pg_database;
db_size_gb
-----
        6252
(1 row)

```

pg_catalog_size

```

prodapgib=> SELECT nspname ,
prodapgib-> pg_size_pretty(sum(pg_relation_size(c.oid))) AS size
prodapgib-> FROM pg_class c
prodapgib-> LEFT JOIN pg_namespace n ON (n.oid = c.relnamespace)
prodapgib-> WHERE nspname IN ('pg_catalog', 'information_schema')
prodapgib-> GROUP by nspname
prodapgib-> ORDER BY nspname DESC
prodapgib-> ;
 nspname | size
-----+-----
pg_catalog | 171 MB
information_schema | 88 kB
(2 rows)

```

pg_db_settings

```

prodapgib=> SELECT name, setting
prodapgib-> FROM pg_settings
prodapgib-> WHERE name IN ('auto_explain.log_analyze','auto_explain.log_buffers','auto_explain.
log_timing','auto_explain.log_min_duration', 'random_page_cost', 'log_statement','shared_preload_libraries',
'work_memory', 'idle_in_transaction_session_timeout') order by 1;

```

name	setting
auto_explain.log_analyze	on
auto_explain.log_buffers	on
auto_explain.log_min_duration	1000
auto_explain.log_timing	on
idle_in_transaction_session_timeout	86400000
log_statement	ddl
random_page_cost	1
shared_preload_libraries	rdsutils,pg_stat_statements,auto_explain,pg_hint_plan,apg_plan_mgmt

(8 rows)

pg_vacuum_settings

```

prodapgib=> -- auto vacuum setting
prodapgib=> select name, setting, source
prodapgib-> from pg_settings
prodapgib-> where name in ('autovacuum', 'autovacuum_freeze_max_age');

```

name	setting	source
autovacuum	on	configuration file
autovacuum_freeze_max_age	200000000	default

(2 rows)

```

prodapgib=> -- current max age
prodapgib=> select max(age(datfrozenxid))
prodapgib-> from pg_database;

```

max
157307062

(1 row)

pg_vacuum_efficiency

```

prodapgib=> SELECT current_database() shard_id, nspname, relname, relpages tot_pages, all_visible visible_pages,
all_visible/relpages::float visibility_factor
FROM (SELECT relname,relpages,nspname, (select all_visible from pg_visibility_map_summary
((nspname||'.'||relname)::regclass) )
prodapgib-> FROM (SELECT relname,relpages,nspname, (select all_visible from pg_visibility_map_summary
((nspname||'.'||relname)::regclass) )
prodapgib(> FROM pg_class c LEFT JOIN
prodapgib(> pg_namespace n ON (n.oid = c.relnamespace)
prodapgib(> WHERE relkind = 'r'
prodapgib(> ) x
prodapgib-> WHERE all_visible > 0
prodapgib-> AND all_visible/relpages::float <= 0.96
prodapgib-> ORDER BY visibility_factor desc ;

```

shard_id	nspname	relname	tot_pages	visible_pages	visibility_factor
prodapgib perfstat	host_stat_hist		12330	11742	
0.9523114355231144					
prodapgib perfstat	pg_stat_activity_sample_p2023_10_06		44559	42402	
0.9515922709216993					

prodapgib perfstat 0.9406830299687442	pg_stat_activity_sample_p2023_10_11	43512	40931
prodapgib perfstat 0.9387721330465001	pg_stat_activity_sample_p2023_11_07	24172	22692
prodapgib perfstat 0.9384905068256708	pg_stat_activity_sample_p2023_11_17	25492	23924
prodapgib perfstat 0.9333759641900803	pg_stat_activity_sample_p2023_11_02	25021	23354
prodapgib perfstat 0.9266091510602978	pg_stat_activity_sample_p2023_09_25	42771	39632
prodapgib perfstat 0.9243072718869934	pg_stat_activity_sample_p2023_10_16	49767	46000
prodapgib perfstat 0.9214353346767341	pg_stat_activity_sample_p2023_10_07	41189	37953
prodapgib perfstat 0.9162358222115347	pg_stat_activity_sample_p2023_10_02	74853	68583
prodapgib perfstat 0.9160941586748038	pg_stat_activity_sample_p2023_10_24	28675	26269
prodapgib perfstat 0.8943111666796906	pg_stat_activity_sample_p2023_11_18	25594	22889
prodapgib perfstat 0.8933502028057902	pg_stat_activity_sample_p2023_11_14	26873	24007
prodapgib ibobadm 0.8928748875970614	psp_source_system_transmission_m102023_from_ews	2127391	1899494
prodapgib perfstat 0.8925574407270329	pg_stat_activity_sample_p2023_11_13	27289	24357
prodapgib perfstat 0.8923758605536839	pg_stat_activity_sample_p2023_11_12	27308	24369
prodapgib perfstat 0.891841165863173	pg_stat_activity_sample_p2023_11_16	27173	24234
prodapgib perfstat 0.8907395069953364	pg_stat_activity_sample_p2023_11_15	27018	24066
prodapgib perfstat 0.8904894671623296	pg_stat_activity_sample_p2023_11_19	25824	22996
prodapgib perfstat 0.8903049783226192	pg_stat_activity_sample_p2023_11_06	26756	23821
prodapgib perfstat 0.8899555226093403	pg_stat_activity_sample_p2023_11_11	26980	24011
prodapgib perfstat 0.8895820103843711	pg_stat_activity_sample_p2023_11_09	26771	23815
prodapgib perfstat 0.8894027589506419	pg_stat_activity_sample_p2023_11_03	25082	22308
prodapgib perfstat 0.8884031206422192	pg_stat_activity_sample_p2023_11_20	26533	23572
prodapgib perfstat 0.8878589832212657	pg_stat_activity_sample_p2023_10_29	31826	28257
prodapgib perfstat 0.8878114604908053	pg_stat_activity_sample_p2023_10_28	31866	28291
prodapgib perfstat 0.8790344827586207	pg_stat_activity_sample_p2023_09_28	43500	38238
prodapgib perfstat 0.8786856212904461	pg_stat_activity_sample_p2023_10_14	41784	36715
prodapgib ibobadm 0.875	psp_hcm401k_company_policy	24	21
prodapgib perfstat 0.8744855190121965	pg_stat_activity_sample_p2023_11_21	26483	23159
prodapgib perfstat 0.8738885107690957	pg_stat_activity_sample_p2023_10_12	45547	39803
prodapgib perfstat 0.8721062063416597	pg_stat_activity_sample_p2023_11_08	27343	23846
prodapgib perfstat 0.8710989034755715	pg_stat_activity_sample_p2023_10_13	46237	40277
prodapgib perfstat 0.8587264413962307	pg_stat_activity_sample_p2023_10_15	44835	38501
prodapgib perfstat 0.8577721837633732	pg_stat_activity_sample_p2023_10_25	31780	27260
prodapgib perfstat 0.8571326387067547	pg_stat_activity_sample_p2023_09_30	41941	35949
prodapgib ibobadm 0.8500322136120407	psp_source_system_transmission_m112023_from_psp	223508	189989
prodapgib pg_catalog	pg_shdepend	180	153

0.85				
prodapgib perfstat	pg_stat_activity_sample_p2023_11_05	28729	24407	
0.8495596783737687				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_18	53355	45295	
0.8489363695998501				
prodapgib perfstat	pg_stat_activity_sample_p2023_11_10	25982	22035	
0.8480871372488646				
prodapgib perfstat	pg_stat_activity_sample_p2023_11_04	26995	22819	
0.8453046860529728				
prodapgib perfstat	pg_stat_activity_sample_p2023_09_29	43103	36409	
0.8446975848548824				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_05	43139	36432	
0.8445258350912168				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_21	47498	39821	
0.8383721419849257				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_01	41916	35125	
0.8379854947991221				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_03	64482	53994	
0.8373499581278496				
prodapgib perfstat	pg_stat_activity_sample_p2023_09_26	42982	35989	
0.837303987715788				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_08	39995	33438	
0.8360545068133517				
prodapgib perfstat	pg_stat_activity_sample_p2023_10_22	47739	39896	
0.8357108443829993				
prodapgib perfstat	pg_stat_activity_sample_p2023_11_01	29120	24197	
0.8309409340659341				
prodapgib perfstat	pg_stat_activity_sample_p2023_09_27	44197	36671	
0.8297169491141932				
prodapgib ibobadm	psp_source_system_transmission_m112023_from_qbdt	705069	584601	
0.8291401267110028				
prodapgib ibobadm	psp_source_system_transmission_m102023_from_qbdt	999911	825885	
0.8259585103074174				
prodapgib ibobadm	psp_source_system_transmission_m112023_from_ews	1216662	991764	
0.8151516197596375				
prodapgib ibobadm	psp_source_system_transmission_m102023_from_psp	345197	267309	
0.7743665211458964				
prodapgib pg_catalog	pg_statistic	5275	3644	
0.6908056872037914				
prodapgib pg_catalog	pg_index	269	151	
0.5613382899628253				
prodapgib pg_catalog	pg_authid	2	1	
0.5				
prodapgib ibobadm	psp_hcm401k_employee_deduction	421	194	
0.46080760095011875				
(60 rows)				